

Chapter 23: Combinatorics

Overview

Combinatorics (Tổ hợp) is the branch of mathematics concerned with counting, arranging, and selecting objects. This chapter introduces fundamental counting principles and formulas that form the foundation for probability theory and many real-world applications.

What you'll master:

- **Counting Principles:** Addition and multiplication rules for systematic counting
- **Permutations:** Arrangements where order matters
- **Arrangements:** Selecting and ordering subsets of objects
- **Combinations:** Selections where order doesn't matter

Real-world connections:

- Password security and cryptography
- Tournament scheduling and bracket design
- Probability calculations in games and statistics
- Resource allocation and optimization problems

Vocabulary & Terminology

Core Terms

English	IPA	Vietnamese	Definition
Combinatorics	/ˌkɒmbɪnəˈtɔːrɪks/	Tổ hợp	The mathematics of counting and arrangement
Permutation	/ˌpɜːmjʊˈteɪʃən/	Hoán vị	An arrangement of objects in a specific order
Arrangement	/əˈreɪndʒmənt/	Chỉnh hợp	A selection of objects arranged in order

Combination	/ˌkɒmbɪˈneɪʃən/	Tổ hợp	A selection of objects without regard to order
Factorial	/fækˈtɔːriəl/	Giai thừa	The product $n! = n \times (n-1) \times \dots \times 2 \times 1$
Counting principle	/ˈkaʊntɪŋ ˈprɪnsəpəl/	Quy tắc đếm	Fundamental rules for systematic counting
Addition rule	/əˈdɪʃən ruːl/	Quy tắc cộng	Count total when actions are mutually exclusive
Multiplication rule	/ˌmʌltɪplɪˈkeɪʃən ruːl/	Quy tắc nhân	Count total when actions occur in sequence
Tree diagram	/triː ˈdaɪəgræm/	Sơ đồ hình cây	Visual representation of sequential choices
Sample space	/ˈsɑːmpəl speɪs/	Không gian mẫu	The set of all possible outcomes

Advanced Terms

English	IPA	Vietnamese	Definition
Circular permutation	/ˈsɜːkjələr ˌpɜːmjʊˈteɪʃən/	Hoán vị vòng tròn	Arrangements of objects in a circle
Permutation with repetition	/ˌpɜːmjʊˈteɪʃən wɪð ˌrepiˈtɪʃən/	Hoán vị lặp	Arrangements with repeated elements
Binomial coefficient	/baɪˈnəʊmiəl ˌkəʊɪˈfɪʃənt/	Hệ số nhị thức	The value C_n^k in binomial expansion

Partition	/pɑ:ˈtɪʃən/	Phân hoạch	Division of a set into disjoint subsets
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Theory

1. Counting Principles (Quy tắc đếm)

1.1 Addition Rule (Quy tắc cộng)

Principle: If an action can be performed in

m

ways OR another action can be performed in

n

ways (and the two actions cannot occur simultaneously), then there are

$m + n$

ways to perform either action.

Mathematical statement:

$$|A \cup B| = |A| + |B| \quad \text{when } A \cap B = \emptyset$$

Example: A student can travel from home to school by bus (3 routes) OR by bicycle (2 routes). Total ways =

$$3 + 2 = 5$$

ways.

1.2 Multiplication Rule (Quy tắc nhân)

Principle: If an action consists of

k

steps performed in sequence, where step 1 can be done in

n_1

ways, step 2 in

n_2

ways, ..., and step

k

in

n_k

ways, then the entire action can be performed in:

$n_1 \times n_2 \times \cdots \times n_k$ ways

Example: A 3-digit PIN code where each digit can be 0-9 has
 $10 \times 10 \times 10 = 1000$
possible codes.

1.3 Tree Diagrams (Sơ đồ hình cây)

Tree diagrams provide a visual method for systematic counting, especially useful when the number of choices at each step is small.

Structure:

- Each branch represents a choice
 - Paths from root to leaves represent complete outcomes
 - Total outcomes = number of leaf nodes
-

2. Permutations (Hoán vị)

2.1 Permutations of n Distinct Objects

Definition: A permutation of

n

distinct objects is an arrangement of all

n

objects in a specific order.

Formula:

$$P_n = n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$$

Convention:

$$0! = 1$$

Example: The number of ways to arrange 5 books on a shelf is:

$$P_5 = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

2.2 Circular Permutations (Hoán vị vòng tròn)

When arranging

n

objects in a circle, rotations are considered identical.

Formula:

$$P_{\text{circular}} = (n - 1)!$$

Example: Seating 6 people around a circular table:

$$(6 - 1)! = 5! = 120$$

ways.

3. Arrangements (Chỉnh hợp)

3.1 Arrangements of k Objects from n

Definition: An **arrangement** (or k-permutation) of

k

objects selected from

n

distinct objects is an ordered selection where order matters.

Formula:

$$A_n^k = \frac{n!}{(n-k)!} = n \times (n - 1) \times \dots \times (n - k + 1)$$

Conditions:

$$1 \leq k \leq n$$

Example: Selecting and arranging 3 students from a class of 20 for president, vice-president, and secretary:

$$A_{20}^3 = \frac{20!}{17!} = 20 \times 19 \times 18 = 6840$$

3.2 Key Properties

1. $A_n^n = n!$
(arrangement of all objects = permutation)
 2. $A_n^1 = n$
(selecting one object)
 3. $A_n^0 = 1$
(empty arrangement)
-

4. Combinations (Tổ hợp)

4.1 Combinations of k Objects from n

Definition: A combination of

k

objects selected from

n

distinct objects is a selection where order does NOT matter.

Formula:

$$C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{A_n^k}{k!}$$

Conditions:

$$0 \leq k \leq n$$

Example: Choosing 3 students from a class of 20 for a committee:

$$C_{20}^3 = \frac{20!}{3! \times 17!} = \frac{20 \times 19 \times 18}{3 \times 2 \times 1} = 1140$$

4.2 Key Properties

1. **Symmetry:**

$$C_n^k = C_n^{n-k}$$

2. **Boundary values:**

$$C_n^0 = C_n^n = 1$$

3. **Pascal's identity:**

$$C_n^k = C_{n-1}^{k-1} + C_{n-1}^k$$

4. **Relationship with arrangements:**

$$A_n^k = k! \times C_n^k$$

5. Distinguishing Permutations, Arrangements, and Combinations

Concept	Order matters?	All objects used?	Formula	Example
Permutation	✓ Yes	✓ Yes	$n!$	Arranging 5 books
Arrangement	✓ Yes	✗ No (select k from n)	$A_n^k = \frac{n!}{(n-k)!}$	Top 3 from 20 students
Combination	✗ No	✗ No (select k from n)	$C_n^k = \frac{n!}{k!(n-k)!}$	Committee of 3 from 20

Decision flowchart:

1. Are we using all objects? → Yes: **Permutation** ($n!$)
2. Does order matter? → Yes: **Arrangement** (A_n^k) | No: **Combination** (C_n^k)

💡 Worked Examples

Example 1: Addition and Multiplication Rules

Problem: A restaurant offers:

- Main courses: 4 meat dishes, 3 vegetarian dishes
- Desserts: 5 options

How many ways can a customer order:

- a) A main course (meat OR vegetarian)?
- b) A complete meal (main course AND dessert)?

Solution:

- a) **Addition rule** (mutually exclusive choices):

Total main courses = $4 + 3 = 7$ ways

b) **Multiplication rule** (sequential choices):

Complete meals = $7 \times 5 = 35$ ways

Answer: a) 7 ways; b) 35 ways

Example 2: Permutations

Problem: In how many ways can 6 students stand in a line for a photo?

Solution:

This is a permutation of 6 distinct objects:

$$P_6 = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

Answer: 720 ways

Example 3: Circular Permutations

Problem: In how many ways can 5 people sit around a circular table?

Solution:

For circular arrangements, we fix one person's position to eliminate rotational duplicates:

$$P_{\text{circular}} = (5 - 1)! = 4! = 24$$

Answer: 24 ways

Example 4: Arrangements

Problem: A school needs to select 3 students from a class of 15 for the positions of class president, vice-president, and secretary. How many ways can this be done?

Solution:

Order matters (different positions), and we're selecting 3 from 15:

$$A_{15}^3 = \frac{15!}{(15-3)!} = \frac{15!}{12!} = 15 \times 14 \times 13 = 2730$$

Answer: 2730 ways

Example 5: Combinations

Problem: A committee of 4 students must be chosen from a group of 12. How many different committees are possible?

Solution:

Order doesn't matter (all committee members have equal status):

$$C_{12}^4 = \frac{12!}{4! \times 8!} = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = \frac{11880}{24} = 495$$

Answer: 495 committees

Example 6: Mixed Problem

Problem: A team of 5 players is to be selected from 8 men and 6 women. How many teams can be formed if the team must include:

- a) Exactly 3 men and 2 women?
- b) At least 3 men?

Solution:

a) **Exactly 3 men and 2 women:**

- Choose 3 men from 8:

$$C_8^3 = \frac{8!}{3! \times 5!} = 56$$

- Choose 2 women from 6:

$$C_6^2 = \frac{6!}{2! \times 4!} = 15$$

- Total teams:

$$56 \times 15 = 840$$

b) **At least 3 men** means 3, 4, or 5 men:

- **3 men, 2 women:**

$$C_8^3 \times C_6^2 = 56 \times 15 = 840$$

- **4 men, 1 woman:**

$$C_8^4 \times C_6^1 = 70 \times 6 = 420$$

- **5 men, 0 women:**

$$C_8^5 \times C_6^0 = 56 \times 1 = 56$$

- Total:

$$840 + 420 + 56 = 1316$$

Answer: a) 840 teams; b) 1316 teams

Applications

1. Cryptography and Password Security

Problem: A password must contain exactly 8 characters: 5 letters (A-Z) and 3 digits (0-9), in any order. How many such passwords exist?

Solution:

- Choose 5 positions for letters from 8 positions:

$$C_8^5 = 56$$

- Fill 5 letter positions:

$$26^5$$

ways

- Fill 3 digit positions:

$$10^3$$

ways

- Total:

$$56 \times 26^5 \times 10^3 = 56 \times 11,881,376 \times 1000 \approx 6.65 \times 10^{11}$$

Insight: Adding complexity requirements dramatically increases password space, improving security.

2. Tournament Scheduling

Problem: In a round-robin tournament with 10 teams (each team plays every other team once), how many games must be scheduled?

Solution:

Each game involves choosing 2 teams from 10, and order doesn't matter:

$$C_{10}^2 = \frac{10!}{2! \times 8!} = \frac{10 \times 9}{2} = 45$$

Answer: 45 games

3. Quality Control Sampling

Problem: A shipment contains 100 items, of which 5 are defective. If a quality inspector randomly selects 10 items for testing, how many different samples are possible?

Solution:

Order doesn't matter in sampling:

$$C_{100}^{10} = \frac{100!}{10! \times 90!} \approx 1.73 \times 10^{13}$$

Extension: The probability that the sample contains exactly 2 defective items is:

$$P = \frac{C_5^2 \times C_{95}^8}{C_{100}^{10}}$$

4. Seating Arrangements with Constraints

Problem: 7 people (including 2 who must sit together) are to be seated in a row. How many arrangements are possible?

Solution:

Method: Treat the 2 people who must sit together as a single "unit":

- We now have 6 units to arrange:
 $6! = 720$
ways
- Within their unit, the 2 people can swap:
 $2! = 2$
ways
- Total:
 $720 \times 2 = 1440$

Answer: 1440 arrangements

Summary

Key Formulas

Concept	Formula	When to Use
Addition Rule	$m + n$	Mutually exclusive choices (OR)
Multiplication Rule	$n_1 \times n_2 \times \cdots \times n_k$	Sequential choices (AND)
Permutation	$P_n = n!$	Arrange all n objects
Circular Permutation	$(n - 1)!$	Arrange n objects in a circle
Arrangement	$A_n^k = \frac{n!}{(n-k)!}$	Select and order k from n
Combination	$C_n^k = \frac{n!}{k!(n-k)!}$	Select k from n (order doesn't matter)

Problem-Solving Strategy

1. **Identify the type:** Does order matter? Are we using all objects?
2. **Check for constraints:** Restrictions, groupings, or special conditions?
3. **Choose the right formula:** Permutation, arrangement, or combination?
4. **Break complex problems:** Use multiplication/addition rules to combine simpler counts
5. **Verify:** Does the answer make intuitive sense?

Common Pitfalls

- Confusing arrangements (order matters) with combinations (order doesn't matter)
- Forgetting to account for constraints (e.g., "at least," "at most")
- Miscounting in problems with multiple stages
- Not recognizing when to use circular permutations

Review Questions

Conceptual Questions

1. Explain the difference between A_n^k and C_n^k using a real-world example.
2. Why is $(n - 1)!$ used for circular permutations instead of $n!$?
3. When should you use the addition rule versus the multiplication rule?
4. Prove that $C_n^k = C_n^{n-k}$ using the combination formula.

Computational Problems

5. Calculate:
 - a) $7!$
 - b) A_{10}^3
 - c) C_{15}^5
6. A 4-digit PIN code is formed using digits 0-9 without repetition. How many PINs are possible?
7. From a deck of 52 cards, how many 5-card hands contain exactly 3 aces?
8. In how many ways can 8 people be seated around a circular table if 2 specific people must NOT sit next to each other?

Application Problems

9. A committee of 6 is to be chosen from 10 men and 8 women. How many committees have:
- a) Equal numbers of men and women?
 - b) At least 4 women?
10. A password consists of 3 letters followed by 4 digits. How many passwords are possible if:
- a) Repetition is allowed?
 - b) Repetition is not allowed?
-

Memory Aids & Tips

Mnemonics

"Order Matters = Arrangement"

- If you care about WHO is president vs. secretary → use A_n^k

- If you just care about WHO is on the committee → use C_n^k

Factorial Growth

- $5! = 120$
(manageable)
- $10! = 3,628,800$
(large!)
- $20! \approx 2.4 \times 10^{18}$
(astronomical!)

Calculator Tips (Casio fx-580VN X)

Computing factorials:

Plain Text

7 SHIFT $\times$! =

→ 5040

Computing arrangements

A_n^k

:

Plain Text

10 SHIFT $\div$ (nPr) 3 =

→ 720

Computing combinations

C_n^k

:

Plain Text

15 SHIFT $\div$ (nCr) 5 =

→ 3003

Visual Comparison

Plain Text

Permutation (all objects, order matters):

ABC, ACB, BAC, BCA, CAB, CBA → 6 ways ($3!$)

Arrangement (select 2 from 3, order matters):

AB, BA, AC, CA, BC, CB → 6 ways (A_3^2)

Combination (select 2 from 3, order doesn't matter):

AB, AC, BC → 3 ways (C_3^2)

Quick Reference: Decision Tree

Plain Text

Do we use ALL objects?

└ YES → Permutation ($n!$)

| └ In a circle? → $(n-1)!$

└ NO → Select k from n

 └ Does ORDER matter?

 └ YES → Arrangement (A_n^k)

 └ NO → Combination (C_n^k)

Conclusion

Combinatorics provides the mathematical foundation for counting and arrangement problems that appear throughout mathematics and its applications. The key to mastery is recognizing problem types and selecting the appropriate counting technique.

In the next chapter (Chapter 24: Probability Basics), we'll use these combinatorial formulas to calculate probabilities of events, connecting counting to the likelihood of outcomes in random experiments.

Practice recommendation: Work through 10-15 problems of each type (permutations, arrangements, combinations) to build pattern recognition and computational fluency.